

# 4.10.18

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## Question:

Find the area of the region bounded by the lines  $y = 4x + 5$ ,  $y = 5 - x$ , and  $4y = x + 5$ .

## Solution:

The bounded region is a triangle. Its area can be determined by first finding the coordinates of its three vertices, which are the points of intersection of the given lines.

### Step 1: Calculation of Vertices

**Vertex A:** Intersection of  $L_1 : -4x + y = 5$  and  $L_2 : x + y = 5$

$$\left( \begin{array}{cc|c} -4 & 1 & 5 \\ 1 & 1 & 5 \end{array} \right) \xrightarrow{4\rho_2 + \rho_1} \left( \begin{array}{cc|c} -4 & 1 & 5 \\ 0 & 5 & 25 \end{array} \right) \quad (1)$$

From the second row,  $5y = 25 \implies y = 5$ . Substituting into the first row,  $-4x + 5 = 5 \implies x = 0$ . The coordinates for Vertex A are  $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ .

**Vertex B:** Intersection of  $L_2 : x + y = 5$  and  $L_3 : -x + 4y = 5$

$$\left( \begin{array}{cc|c} 1 & 1 & 5 \\ -1 & 4 & 5 \end{array} \right) \xrightarrow{\rho_1 + \rho_2} \left( \begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 5 & 10 \end{array} \right) \quad (2)$$

From the second row,  $5y = 10 \implies y = 2$ . Substituting into the first row,  $x + 2 = 5 \implies x = 3$ . The coordinates for Vertex B are  $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ .

**Vertex C:** Intersection of  $L_1 : -4x + y = 5$  and  $L_3 : -x + 4y = 5$

$$\left( \begin{array}{cc|c} -4 & 1 & 5 \\ -1 & 4 & 5 \end{array} \right) \xrightarrow{-4\rho_2 + \rho_1} \left( \begin{array}{cc|c} -4 & 1 & 5 \\ 0 & -15 & -15 \end{array} \right) \quad (3)$$

From the second row,  $-15y = -15 \implies y = 1$ . Substituting into the first row,  $-4x + 1 = 5 \implies x = -1$ . The coordinates for Vertex C are  $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ .

### Step 2: Calculation of Area

The area of a triangle with vertices **A**, **B**, and **C** can be calculated using the determinant of the vectors forming two of its sides. The formula is:

$$\text{Area} = \frac{1}{2} |\det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C})|$$

The given vertices are represented by 2D vectors:

$$\mathbf{A} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

First, we find the two side vectors:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} \quad (4)$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad (5)$$

Next, we calculate the determinant of the matrix formed by these two vectors as columns:

$$\begin{aligned} \det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C}) &= \left| \begin{pmatrix} -3 & 1 \\ 3 & 4 \end{pmatrix} \right| \\ &= (-3)(4) - (1)(3) \\ &= -12 - 3 = -15 \end{aligned} \quad (6)$$

The area is half the absolute value of this determinant:

$$\text{Area} = \frac{1}{2} |-15| = 7.5 \quad (7)$$

Thus, the area of the triangle is 7.5 square units.

